

1) Concise summary (examiner's words)

Problem & goal. You study how to **route** mixed clinical-style inputs (tabular biomarkers and free-text queries) to the **right expert module**—Prediction, Case-Based Reasoning (CBR), QA, or Summarisation—under **tight latency/memory budgets**, with **calibrated uncertainty** and the ability to **abstain/hedge**. The router should be lightweight ($p_{95} \leq 100$ ms, ≤ 2 GB) and improve end-to-end utility vs single-module baselines.

Core contributions.

- A **text-only, temperature-calibrated** routing policy with abstention/hedging; optional LLM-intent prior mixed in logit space.
- **Four-module orchestration** with consistent interfaces + explanations (SHAP for Prediction, prototype/nearest-neighbour rationales for CBR, evidence-grounded citations for QA).
- Full **reproducibility**: fixed splits, seeds, calibration files, indices, checkpoints, and JSONL decision logs.

Data & setup.

- **Router corpus**: 10,240 labelled queries (train 6,656 / val 2,048 / test 1,536, balanced).
- **Prediction**: synthetic 100k (74 vars, 3 classes) + hybrid 2,895 (78 vars, class imbalance).
- **CBR**: 10,000 cases (70 numeric + de-identified narrative) → 8k index / 2k eval.
- **QA/Sum**: 30,636 text passages, pre-chunked/embedded; retrieval only at eval time. All preprocessing on train to prevent leakage.

Methodology (signals & policies).

- **Router**: query ($\pm$ short context) embeddings + interaction features → small MLP head → **temperature scaling**; thresholds for **abstention**; optional **intent prior**; **LinTS** optional for online adaptation; **hedging** allowed.
- **Prediction**: XGBoost/MLP with grid searches, calibration; SHAP used for interpretability.
- **CBR**: hybrid fusion of semantic text similarity and **Bayesian Cluster Model** (K-means prototypes + Gaussian approx.) with weighted scoring and fast ANN search; module returns coverage/quality signals to the router.
- **QA/Sum**: hybrid retrieval (BM25+dense) + optional cross-encoder reranker; generation constrained to retrieved evidence; module emits retrieval adequacy and generator-confidence signals.
- A **reproducibility harness** fixes seeds, uses config-driven runs, profiles latency, and bootstraps CIs; artifacts are released with integrity checks.

Key results.

- **Routing** (test $n=1,536$): Accuracy **0.9668** (Wilson 95% CI [0.9566, 0.9747]); **ECE** ↓ after calibration ($\tau \approx 0.992$); router-only p_{50} **18–24 ms**, p_{95} **70–95 ms**; end-to-end p_{95} driven by retrieval/generation, not routing.
 - **Prediction**: Synthetic very high (e.g., Macro-F1 ~ 0.99); **Hybrid** moderate (Macro-F1 ~ 0.70 ; Macro-AUROC ~ 0.77) with wider minority-class CIs; calibration improves thresholding.
 - **CBR**: Ablation (6.1) shows **Text-Only** top-1 100% vs **BCM-re-ranked** 57.81% vs **Unified** 83.88% at ~ 70 ms; measured section 7.3 reports **Top-1 0.78–0.85**, **Top-3 0.88–0.92**, text-path **20–40 ms**.
 - **QA/Sum**: **QA EM 0.45–0.55**, F1 0.55–0.65; **ROUGE-1 0.18–0.25**, -2 0.06–0.10, -L 0.15–0.22; retrieval-confidence gate **reduces unsupported claims by $\sim 40\%$** .
 - **End-to-end** (RQ1): **DLQR** beats Always-Prediction / Always-QA on Macro-F1/AUROC at matched budgets; tail latency mass sits in retrieval/reranking/generation stages.
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2) Likely examiner questions (by theme)

A. General / background

1. What precise gap does **DLQR** fill vs cascades or static routers? (cite p95/2 GB constraints, text-only input).
2. State your **RQ1–RQ4** and how each is tested.
3. Why calibrate probabilities for routing thresholds/abstention?
4. Ethical scope: why software-only, synthetic/hybrid data, and no clinical claims?

B. Methods / design choices

5. Justify **text-only** routing and optional **LLM intent prior**; why not fuse tabular inputs at the gate?
6. Why **temperature scaling** (τ , τ_{mix}) and not alternative calibration?
7. Explain **abstention threshold TU** and **hedging**; when is LinTS used?
8. CBR fusion weighting (α) and why BCM-only re-ranking harms top-1 in ablation.

C. Data / experiments

9. Splits, leakage controls, and class imbalance handling for hybrid biomarkers.
10. Router corpus construction, balance, and why text-plus-context helps (or not).
11. QA/Sum retrieval corpora, chunking and indexing choices; reranker on/off.

D. Results / analysis

12. Interpret router metrics: **accuracy 0.9668**, **ECE drop**, latencies; why tail $\neq$ router.
13. Reconcile **CBR 100% top-1 (Text-Only)** in ablation with **0.78–0.85** in measured section.
14. Prediction: why synthetic $\gg$ hybrid, and implications for deployment.
15. QA safety: effect of retrieval-confidence gate ($\sim 40\%$ fewer unsupported claims).

E. Future work & impact

16. Making calibration robust under **distribution shift**; conformal acceptance layers.
17. Online learning: safe policy updates, data governance, and audit trails.

F. Critical challenges / defenses

18. “Text-only router is too limited”—why is it a **feature** under system constraints?
 19. “End-to-end p95 $\sim$ tens of seconds is high”—attribute tails and justify budgets.
 20. Reproducibility: exactly how would I **rerun** every table/figure? (tag + harness).
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3) Interactive mock viva (15–20 Qs with feedback)

Use this like a **live drill**: answer out loud, then compare to the “Feedback & upgrade” right below each question.

Q1 (motivation & RQs)

In 60–90 seconds, describe the gap your work addresses and list **RQ1–RQ4**.

Feedback & upgrade: You should map (a) mixed-modality inputs → need to route; (b) compute budgets ($p_{95} \leq 100$ ms, ≤ 2 GB) → lightweight, calibrated gate; (c) RQ1–RQ4 verbatim. If you skipped budgets or abstention/hedging, add them. Finish with one sentence on **why text-only** gate.

Q2 (contributions)

What are your **three main contributions**?

Feedback & upgrade: Name (1) calibrated text-only router (+LLM prior), (2) four-module orchestration with explanations, (3) full reproducibility & logs. Tie each to a chapter/section.

Q3 (router design)

Why **temperature scaling** and how does it change ECE/NLL without altering accuracy?

Feedback & upgrade: Define τ , objective, expected effect on ECE/NLL; say thresholds use calibrated posteriors. Mention $\tau \approx 0.992$ / $\tau_{\text{mix}} \approx 1.0$ as fitted on validation.

Q4 (abstention & hedging)

When do you **abstain** and when do you **hedge** to multiple targets?

Feedback & upgrade: Quote TU sweep (0.5–0.8) and hedge policy (top-2 under high uncertainty). Mention LinTS optional for online exploration.

Q5 (router metrics)

Summarise **router** accuracy, calibration, and latency; where do **tails** come from?

Feedback & upgrade: Accuracy 0.9668 (CI...), ECE↓ after calibration; p50 18–24 ms, p95 70–95 ms; tails mainly retrieval/generation. If you conflated router with end-to-end, correct it.

Q6 (prediction module)

Why do results differ between **synthetic** and **hybrid** biomarker sets?

Feedback & upgrade: Synthetic separable by construction → high scores; hybrid sparse/imbalanced → moderate metrics; calibration aids thresholding/deferral. Mention Macro-F1 ~ 0.70 , AUROC ~ 0.77 on hybrid.

Q7 (interpretability)

How do you **explain** outputs across modules?

Feedback & upgrade: SHAP for Prediction; CBR prototypes/nearest neighbours + numeric/text rationale; QA returns citations and enforces evidence-grounded responses. Keep it crisp and module-wise.

Q8 (CBR ablation vs measured)

Explain why **Text-Only** got 100% top-1 in ablation but **0.78–0.85** in measured results.

Feedback & upgrade: Contrast controlled ablation setting vs broader measured setup; note **BCM-only re-rank** degraded accuracy; fusion recovers part of it; emphasize design insight: don’t disrupt strong text neighbours with naïve numeric re-ranking.

Q9 (CBR method)

Describe CBR’s **fusion** and what signals it returns to the router.

Feedback & upgrade: Weighted blend (α) of text similarity and BCM probability; returns aggregate retrieval score + coverage flag.

Q10 (QA safety)

How did you reduce **unsupported claims** in QA?

Feedback & upgrade: Hybrid retrieval + optional cross-encoder; **retrieval-confidence gate** cuts unsupported claims by ~40% with modest recall loss; generation constrained to retrieved evidence.

Q11 (end-to-end RQ1)

What did the **end-to-end** comparisons show vs single-module baselines?

Feedback & upgrade: DLQR wins on Macro-F1/AUROC at matched budgets (list ranges) and keeps tail latency competitive; tails live in retrieval/generation. Reference Table 7.5.

Q12 (protocols & metrics)

Name the main **metrics** and how CIs are computed.

Feedback & upgrade: Accuracy, Macro-F1, AUROC/AUPRC; ECE, Brier, NLL; latency p50/p95; throughput; cost proxy; Wilson for proportions and stratified bootstrap for F1/AUROC/AUPRC (1,000 resamples).

Q13 (reproducibility)

How can I **rerun** your experiments exactly?

Feedback & upgrade: Mention archival git tag and harness (fixed splits, preprocessing on train only, profiler, bootstrap evaluator); artifacts and checksums released. Provide the tag note succinctly.

Q14 (router inputs)

Why restrict the router to **text-only** inputs?

Feedback & upgrade: Simplicity, speed, memory; avoids leaking structured features; modules already specialise; matches production constraint (p95 ≤ 100 ms, ≤ 2 GB).

Q15 (design trade-offs)

How did α (CBR fusion) and w (LLM-prior mix) affect performance?

Feedback & upgrade: Note $\alpha=0.7$ used; $w \in \{0.3, 0.5, 0.7\}$ in ablations; describe stability/accuracy trade-offs; temperature scaling applied post-hoc. Keep to what's reported.

Q16 (threats to validity)

Give two **threats to validity** and your mitigations.

Feedback & upgrade: Synthetic/hybrid data representativeness; distribution shift $\rightarrow$ consider conformal layers; measurement biases handled via fixed splits, seeds, bootstrap CIs, and decision logs.

Q17 (online learning)

How would you update thresholds **safely** over time?

Feedback & upgrade: Log everything; use LinTS with conservative priors; keep rollbackable configs; keep abstention gates strict until evidence accumulates.

Q18 (cost/latency)

Explain the **cost proxy** and how it informed routing.

Feedback & upgrade: Weighted function of CPU ms + token counts; used for sensitivity and tie-breaking to avoid expensive paths when utility marginal.

Q19 (module boundaries)

Why decouple **QA** and **Summarisation**?

Feedback & upgrade: QA must be evidence-grounded; Summarisation is safe fallback for synthesis/rewrites; each reports adequacy/confidence back to router.

Q20 (future work)

What's the most impactful **next step**?

Feedback & upgrade: Distribution-free calibration (conformal acceptance) + more realistic, non-synthetic evaluation; stricter governance while retaining p95/2 GB budget.

4) Overall feedback & improvement tips (scorecard)

Strengths

- Clear **systems framing** with explicit **budgets** and router design; strong **calibration** story.
- Solid **reproducibility harness** and artifact release ethos.
- Coherent **module explanations** and safety guardrails (QA gate, evidence-only gen).

Weaknesses / likely probes

- **Distribution shift** and calibration robustness need explicit plans (bring up conformal acceptance).
- **CBR**: reconcile ablation 100% vs measured 0.78–0.85 clearly; justify α and show why BCM-only hurts.
- **Hybrid prediction**: discuss clinical-adjacent implications of moderate metrics and deferral thresholds.

Delivery tips (to stay within 20–30 minutes)

- Open with a **90-second elevator**: problem → constraints → 3 contributions → headline results.
- For methods, **one sentence per module** plus the router policy and calibration.
- Quote **numbers once** (router 0.9668 acc; p95 ≤ 100 ms; QA EM range; CBR top-1 range).
- Use **structured phrases**: “We calibrated, then thresholded; we hedged when uncertain; we logged every decision.”
- Pre-empt criticisms with “**By design** the router is compute-bounded; tails are in retrieval/generation.”

Indicative scoring (self-check)

- **Understanding (25%)**: Can you state RQs, contributions, and constraints crisply?
- **Methodological rigour (25%)**: Calibration, abstention, protocols, metrics & CIs.
- **Results & interpretation (25%)**: Correct numbers + correct **attribution of latency**.
- **Limitations & ethics (15%)**: Software-only scope; governance; synthetic/hybrid rationale.
- **Clarity & time (10%)**: Tight openings, specific citations, decisive conclusions.